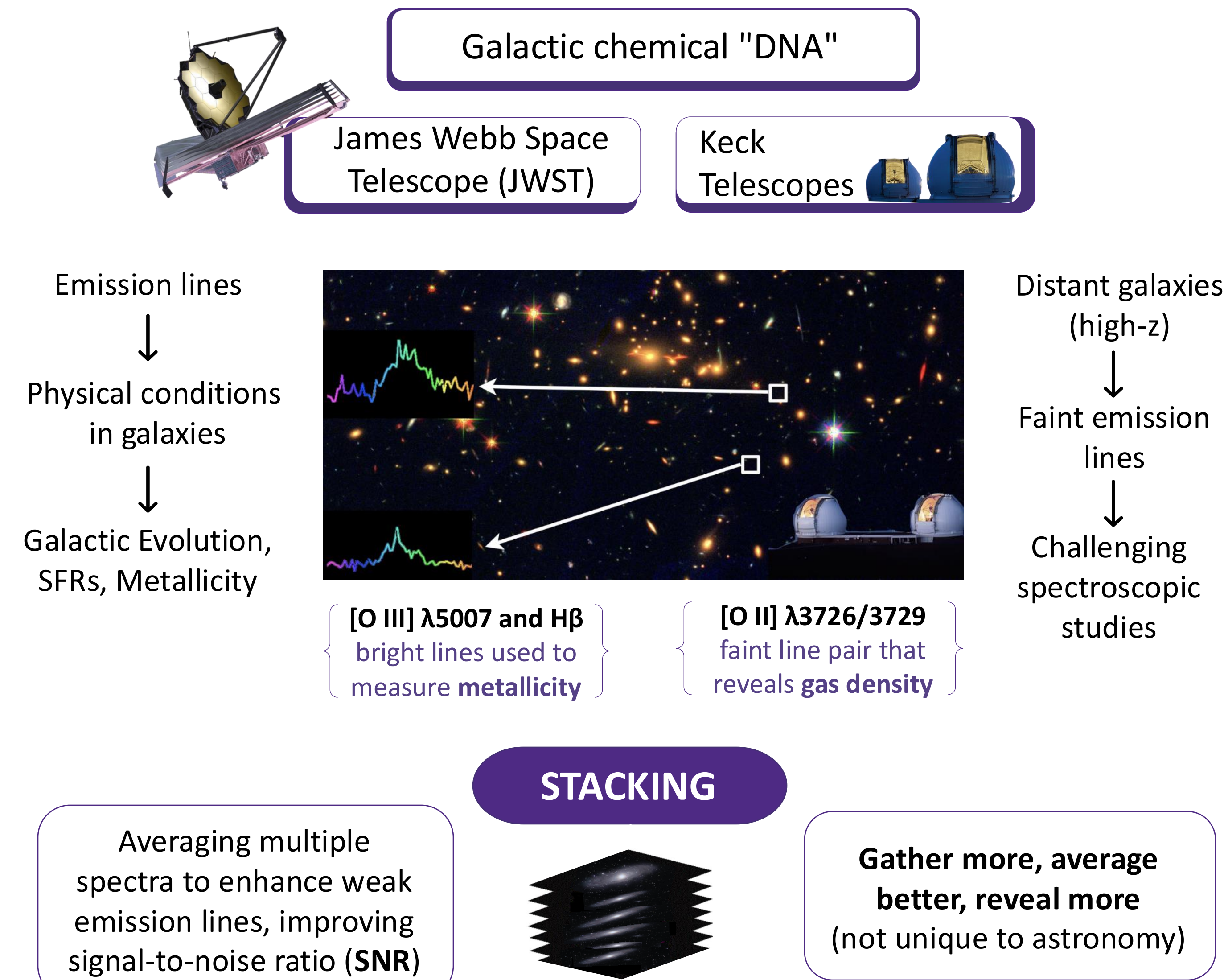


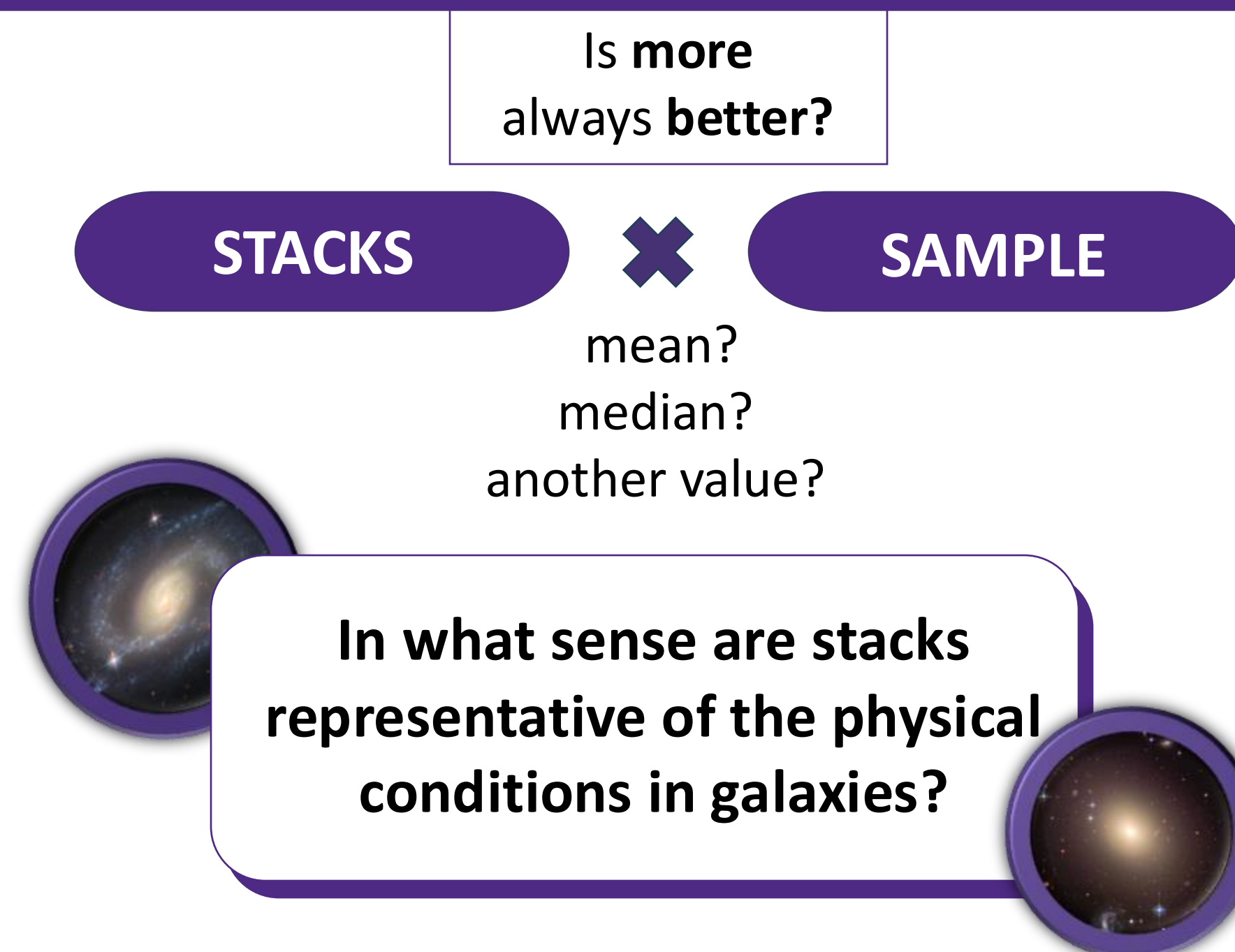
Let Them Shine: Revealing Faint Galaxies with the Spectral Stacking Technique

By Katarina Andrade
mentored by Dr. Allison Strom and Dr. Bryan Scott

Background



Research Goal



References

1. Izotov, Yuri I., et al. "The chemical composition of metal-poor emission-line galaxies in the Data Release 3 of the Sloan Digital Sky Survey." *Astronomy & Astrophysics* 448.3 (2006): 955-970.
2. Nagamine, Kentaro, et al. "Probing Feedback via IGM tomography and the Ly α Forest with Subaru PFS, TMT/ELT, and JWST." *The Astrophysical Journal* 914.1 (2021): 66.
3. Strom, Allison L., et al. "CECILIA: The Faint Emission Line Spectrum of $z \sim 2-3$ Star-forming Galaxies." *The Astrophysical Journal Letters* 958.1 (2023): L11. of *Spatially Extended Ly α Emission in the KBSS-KCWI Survey* (2023, *ApJ*, 953, 118). *The Astrophysical Journal* 961.1 (2024): 143.

Methods

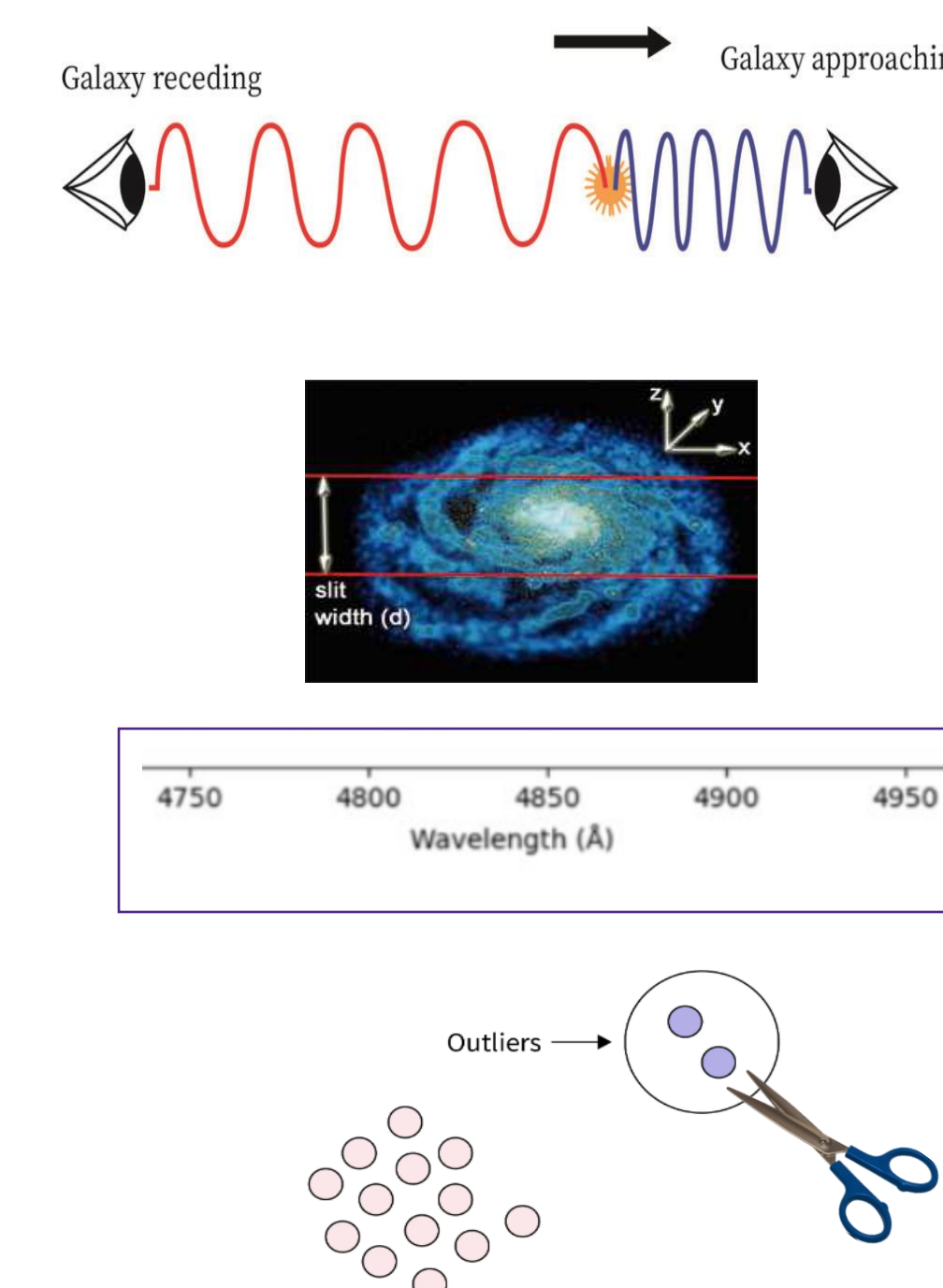
1) Data collection

Keck Baryonic Structure Survey (KBSS)

Chemical Evolution Constrained using Ionized Lines in Interstellar Aurorae (CECILIA)

2) Applying Corrections

"De-redshifting"	Shifts each galaxy's spectrum to its rest frame to enable consistent comparison across the sample
Slit-corrections	Account for instrumental effects that distort measured fluxes for each galaxy
Interpolation	Places all spectra onto a common wavelength grid, necessary for combining fluxes during stacking
Sigma-clipping	Removes extraneous data points (outliers) that are not representative



3) Stacks!

Results

Figure 1. Individual spectra (left) and stacked spectrum (right) for galaxies in the H-band. The individual plots show noisy, weak emission lines. Stacking boosts the SNR, revealing key emission features.

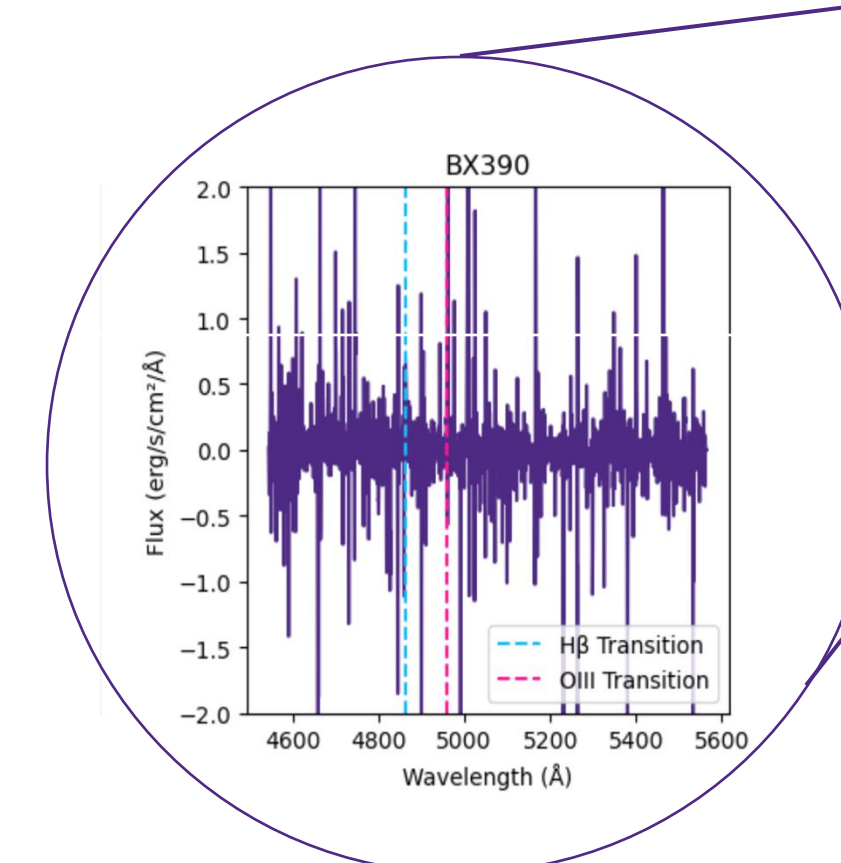
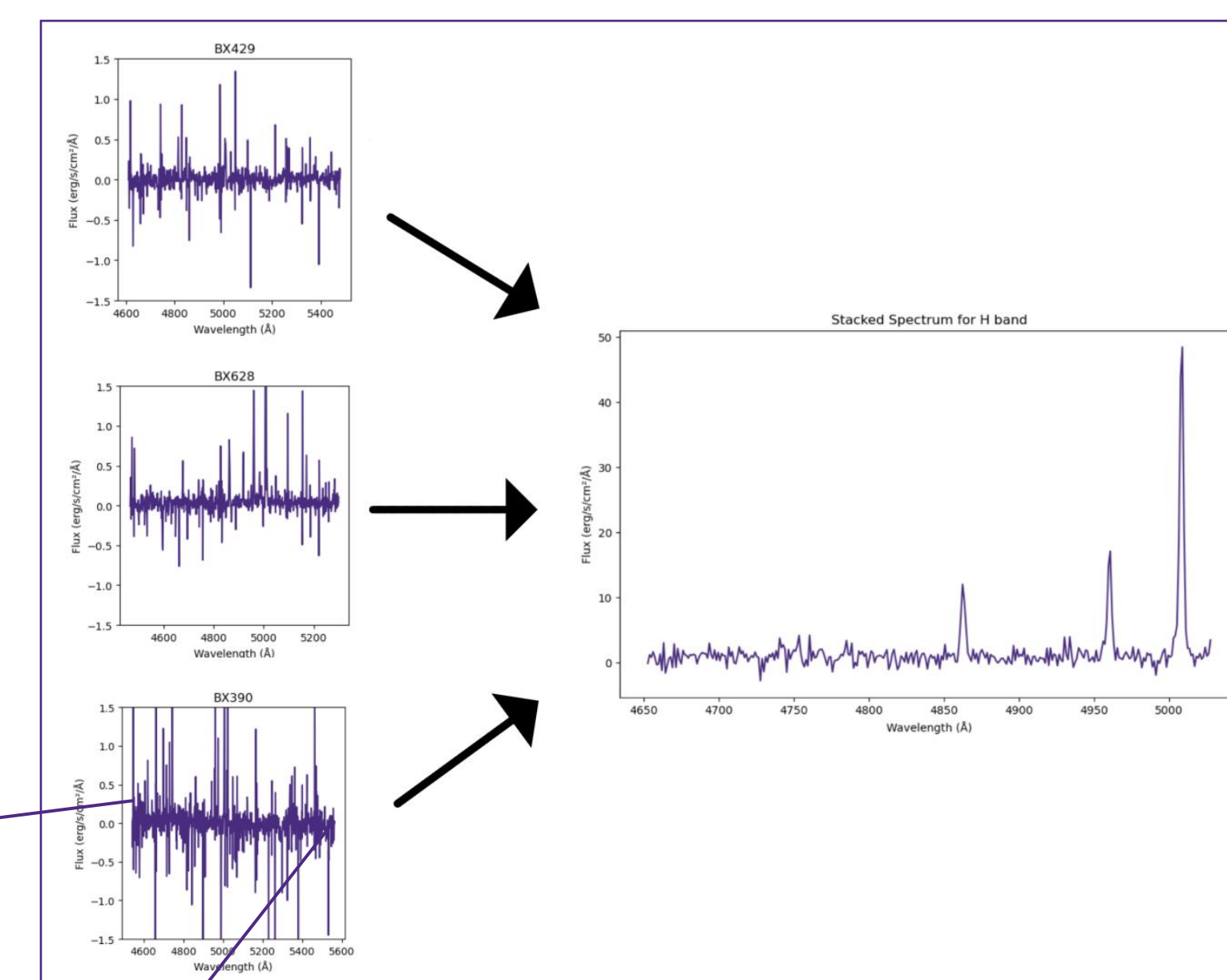


Figure 2. Individual spectrum with emission line markers. Vertical dashed lines highlight the expected positions of the H β and [O III] transitions.

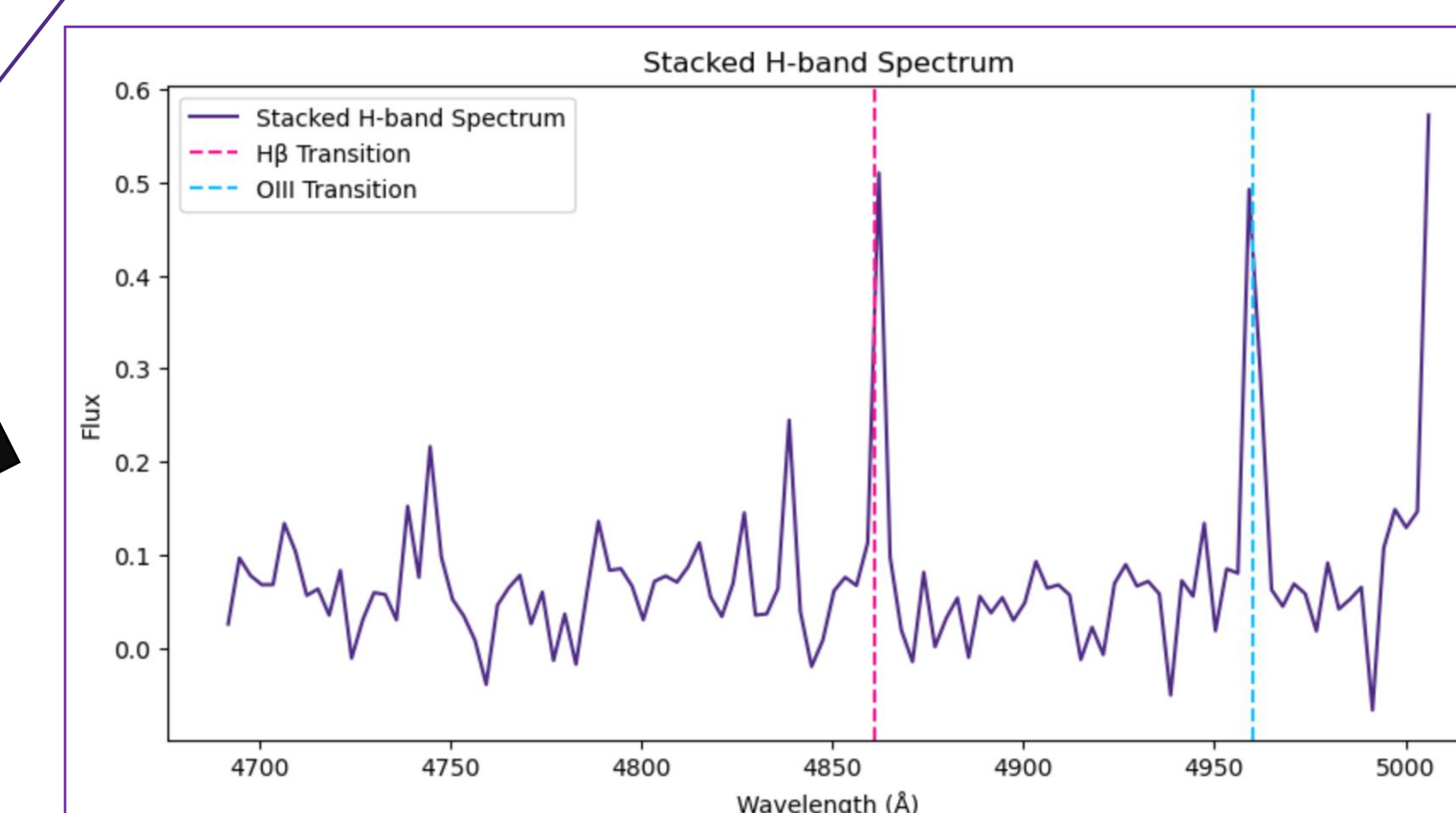


Figure 3. Stacked H-band spectrum with marked emission lines. The stacked spectrum shows clearly resolved emission features at the expected wavelengths of H β and [O III], highlighted by dashed lines.

Conclusion

Spectral Stacking is a powerful tool to advance the study of galaxies

opens new doors for studying faint, high-redshift galaxies

improves SNR, revealing emission lines that would otherwise remain hidden



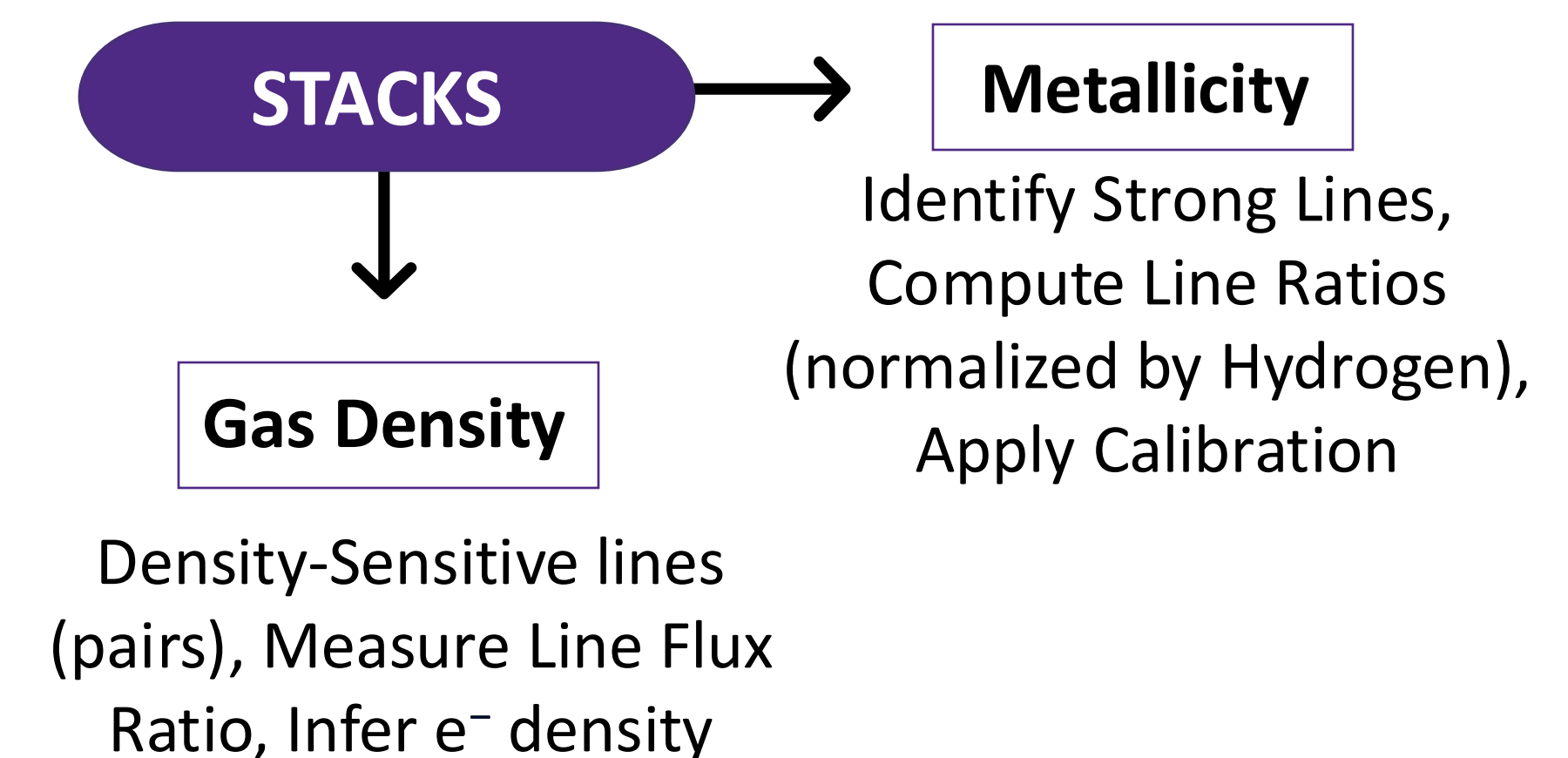
HOWEVER

Is more always better?

the answer might challenge assumptions long held in the field!

not understanding their biases can mislead conclusions

Future Work



Identify any biases introduced by stacking

Acknowledgements

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